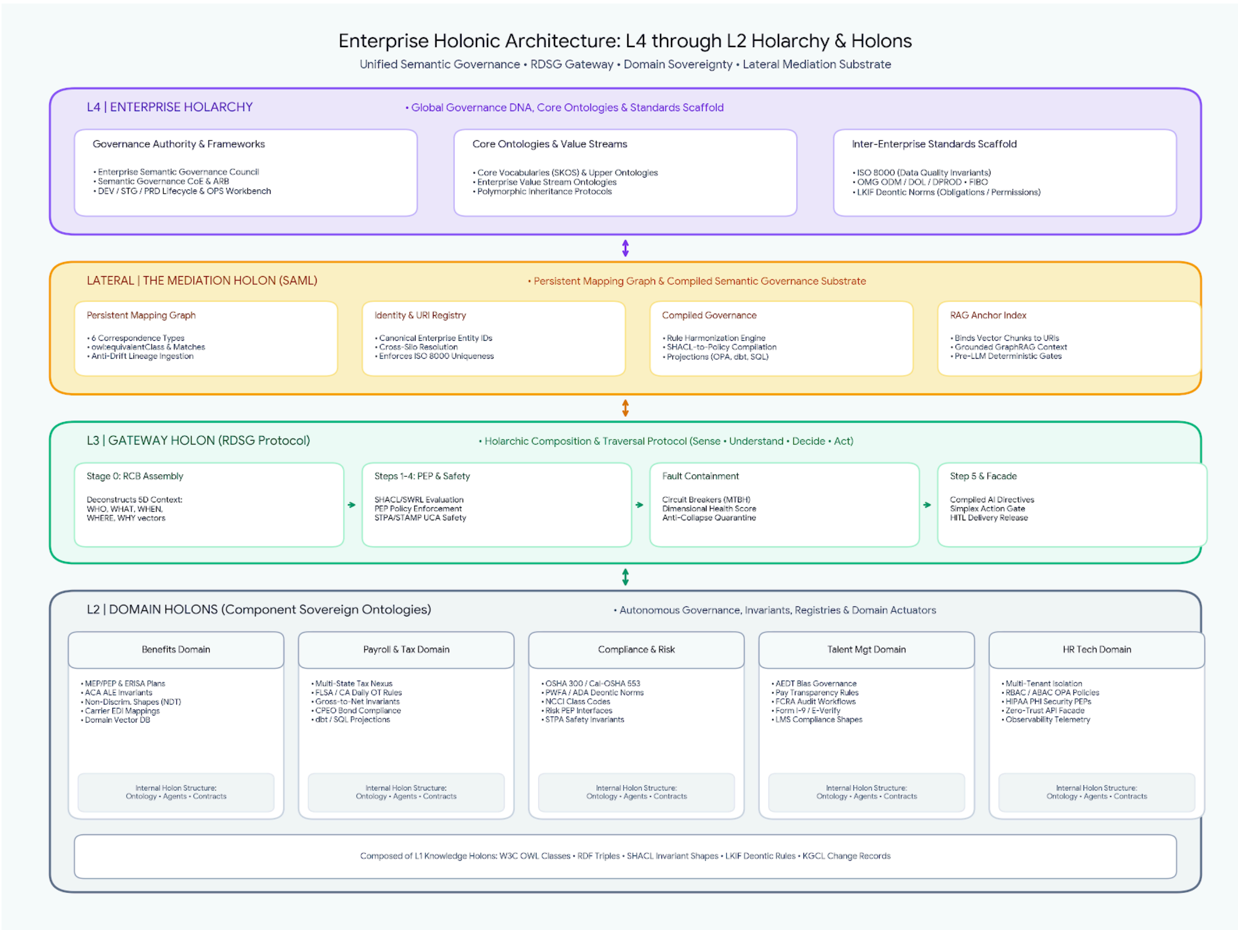
To instantiate regulatory, statutory, and service policies across Insperity's 5 core domains, you implement a **Holonic Semantic Governance Architecture** that separates **Definition-Time canonical truth** (authored in RDF/OWL/LKIF/SHACL) from **Run-Time execution** (projected to OPA/Rego, dbt/SQL, and Agentic guardrails).



The generated architecture diagram illustrates the structural levels from **L4 (Enterprise Holarchy)** down to **L2 (Domain Holons)**, including the lateral **Mediation Holon (SAML)** and underlying **L1 Knowledge Holons**:

L4 — Enterprise Holarchy (Global Governance DNA)

- **Governance Authorities:** The Enterprise Semantic Governance Council and Center of Excellence (CoE) oversee enterprise-wide semantic standards, ARB change reviews, and the DEV/STG/PRD lifecycle.
- **Core Semantic Assets:** Houses foundational vocabularies (SKOS), upper ontologies, and Value Stream Ontologies that propagate polymorphic inheritance rules across all domain sub-holons.

- **Inter-Enterprise Standards Scaffold:** Anchors global invariants and deontic frameworks, including ISO 8000 data quality dimensions, OMG ODM/DOL/DPROD specifications, and LKIF legal norms.

The Lateral Mediation Holon (Semantic Abstraction & Mediation Layer)

- **Persistent Mapping Graph:** Maintains six correspondence types between canonical ontology concepts and physical implementations (dbt, LookML, Neo4j, vector stores) with anti-drift lineage agents.
- **Identity & URI Resolution Registry:** Resolves cross-silo identifiers into canonical Enterprise Entity IDs to enforce the ISO 8000 Uniqueness invariant.
- **Compiled Semantic Governance:** A Rule Harmonization Engine detects constraint conflicts at definition time, while Rule Projection Agents compile SHACL/SWRL invariants down to high-speed runtime policies (OPA/Rego, dbt tests, SQL check constraints).
- **RAG Anchor Index:** Grounds unstructured vector embeddings against canonical URIs before runtime retrieval and governance filtering.

L3 — Gateway Holon (RDSG Semantic Gateway Protocol)

- **Requirement Context Bundle (RCB):** Decomposes inbound requests across five relevancy dimensions (WHO, WHAT, WHEN, WHERE, WHY).
- **Policy Enforcement Points (PEPs):** Enforces compliance, security, risk, and STPA/STAMP safety invariants across domain boundaries.
- **Circuit Breakers & MTBH Monitoring:** Tracks Mean Time Between Hallucinations across relevancy dimensions and isolates fault domains to prevent model collapse.
- **Simplex Action Gates:** Intercepts irreversible or destructive actions for mandatory human authorization before release.

L2 — Domain Holons (Sovereign Component Ontologies)

- **Domain Sovereignty:** Self-contained units for core business services (Benefits, Payroll & Tax, Compliance & Risk, Talent Management, and HR Technology).
- **Internal Anatomy:** Each domain holon encapsulates its own OWL domain ontology, domain vector database, agent registry, data contracts, and compiled SHACL/SWRL invariants—all composed from atomic L1 Knowledge Holons.

Federated Ontology Modularization (L2 Domain Holons)

Define a root Core PEO Ontology (isp-core) and federate the 5 core services into sovereign domain ontologies (L2 Domain Holons) linked via polymorphic inheritance and owl:imports:

- https://insperity.com/ontologies/core# (isp-core): Canonical entity types (ClientCompany, Worker, Jurisdiction, EmploymentType, StatutoryStandard).
- https://insperity.com/ontologies/benefits# (isp-ben): Multiple Employer Plans (MEP/PEP), ERISA plan classes, ACA ALE thresholds, NDT rules.
- https://insperity.com/ontologies/payroll# (isp-pay): Multi-state wage bases, FLSA/state salary basis tests, overtime algorithms, SUTA/SIT nexus.
- https://insperity.com/ontologies/compliance# (isp-comp): Cal/OSHA SB 553, PWFA/PUMP accommodations, OSHA 300 logs, NCCI workers' comp codes.
- https://insperity.com/ontologies/talent# (isp-tal): NYC Local Law 144 AEDT bias rules, state pay scale transparency, FCRA background workflows.
- https://insperity.com/ontologies/hrtech# (isp-tech): Tenant data isolation, RBAC/ABAC models, HIPAA Security/SOC 2 PEP enforcement points.

Design-Time: Representing Rules as Deontic & Invariant Triples

At Design Time, express all statutory policies as **L1 Knowledge Holons** combining **LKIF Deontic statements** (prescribing legal permissions, obligations, and prohibitions) with **W3C SHACL Shapes** (structural invariants).

Code snippet

```
@prefix isp-core: <https://insperity.com/ontologies/core#> .
@prefix isp-pay:  <https://insperity.com/ontologies/payroll#> .
@prefix lkif:     <http://www.estrellaproject.org/lkif-core/norm.owl#> .
@prefix sh:       <http://www.w3.org/ns/shacl#> .
@prefix xsd:      <http://www.w3.org/2001/XMLSchema#> .

# 1. LKIF Deontic Norm: California Daily Overtime Obligation
isp-pay:CADailyOvertimeObligation a lkif:Obligation ;
   lkif:isSubjectTo isp-pay:CaliforniaLaborCodeSection510 ;
   isp-core:governedService isp-core:PayrollService ;
   isp-core:appliesToJurisdiction "US-CA" ;
   isp-core:appliesToWorkerClassification isp-core:NonExempt ;
   isp-core:thresholdHoursDaily "8.0"^^xsd:decimal ;
   isp-core:rateMultiplier "1.5"^^xsd:decimal ;
   isp-core:lifecyclePhase "Run" ;
   isp-core:managedUnderContract "PEO_Client_Allocation_Matrix_Standard" .

# 2. SHACL Invariant Shape: Validating Daily Timesheet Entries
isp-pay:CADailyTimesheetShape a sh:NodeShape ;
```

```
  sh:targetClass isp-pay:TimesheetRecord ;
  sh:property [
    sh:path isp-pay:hoursWorked ;
    sh:datatype xsd:decimal ;
    sh:minInclusive 0.0 ;
  ] ;
  sh:sparql [
    a sh:SPARQLConstraint ;
    sh:message "Non-exempt hours over 8.0/day in CA must apply 1.5x overtime multiplier." ;
    sh:select """
      PREFIX isp-pay: <https://insperity.com/ontologies/payroll#>
      PREFIX isp-core: <https://insperity.com/ontologies/core#>
      SELECT $this
      WHERE {
        $this isp-pay:workedJurisdiction "US-CA" ;
            isp-pay:classification isp-core:NonExempt ;
            isp-pay:hoursWorked ?h ;
            isp-pay:overtimeApplied ?ot .
        FILTER (?h > 8.0 && ?ot != true)
      }
    """ ;
  ] .
```

- **Provenance and Change Management:** Bind every RDF triple change to an **Ontology Requirements Specification Document (ORSD)** and stamp updates with **Knowledge Graph Change Language (KGCL)** metadata tracking *WHO*, *WHAT*, *WHEN*, *WHERE*, and *WHY*.

Design-Time: Rule Harmonization & Cross-Jurisdiction Resolution

When federal, state, and local rules conflict (e.g., federal FLSA exempt salary threshold vs. California/New York thresholds, or federal minimum wage vs. municipal minimum wage), a **Rule Harmonization Engine** executes SPARQL scans over the rule graph[cite: 2]:

Code snippet

```
PREFIX lkif: <http://www.estrellaproject.org/lkif-core/norm.owl#>
PREFIX isp-core: <https://insperity.com/ontologies/core#>

SELECT ?jurisdiction (MAX(?threshold) AS ?effectiveExemptSalaryThreshold)
WHERE {
  ?rule a lkif:Obligation ;[cite: 2]
      isp-core:appliesToJurisdiction ?jurisdiction ;
      isp-core:salaryThresholdValue ?threshold .
}
GROUP BY ?jurisdiction
```

The engine flags overlaps and enforces the statutory principle of **highest employee protection/most stringent rule** before rule publication[cite: 2].

SHACL-to-Policy Compilation Pipeline (Bridging Design Time to Run Time)

To ensure high-throughput processing across thousands of SMB client tenants without running expensive in-graph reasoning on every payroll or benefit transaction, canonical RDF/SHACL invariants pass through a 6-stage compilation pipeline[cite: 2]:

```
[Canonical SHACL/SWRL/LKIF]
        |
        ▼
 1. Shape Parser ──────────────► Parse constraint logic[cite: 2]
 2. Constraint Extractor ──► Extract cardinality, thresholds, deontic norms[cite: 2]
 3. Harmonization Check ────► SPARQL scan for cross-statutory conflicts[cite: 2]
 4. Target Mapper ─────────────► Map to target runtime engine[cite: 2]
 5. Policy Generator ────────► Emit native code (OPA/Rego, dbt/SQL, JSON Schema)[cite: 2]
 6. KGCL Version & Publish ─► Stamp with canonical URI & deploy[cite: 2]
        |
        ▼
[Compiled Policy Registry] ──► Microsecond Execution at L3 Gateway / PEPs[cite: 2]
```

Run-Time Execution: Virtualization & Policy Enforcement Points (PEPs)

At Run Time, the **Virtualization Layer** and **RDSG Gateway Holon** enforce compiled policies at microsecond latency while preserving full URI audit traceability back to the originating LKIF/SHACL triples[cite: 2]:

- **Payroll & Tax Engines (dbt / SQL Projections):** SHACL threshold shapes are projected into dbt data-quality tests and SQL check constraints executed during gross-to-net payroll runs[cite: 2].
- **HR Technology & Multi-Tenant Access (OPA / Rego Projections):**
- Code snippet

```
package insperity.authz

# Compiled from isp-tech:HIPAASecurityShape (URI:
https://insperity.com/ontologies/hrtech#HIPAASecurityShape)[cite: 2]
default allow = false

allow {
   input.action == "READ_PHI"
   input.user.role == "BenefitsAdministrator"
   input.user.tenant_id == input.resource.client_id
```

```
    input.session.mfa_authenticated == true
}
```

-
-
- **Talent & Benefits Agent Guardrails (Compiled AI Directives):** When employee/client AI virtual assistants generate answers regarding leave, 401(k) eligibility, or job descriptions, the RDSG gateway injects compiled JSON Schema constraints (Step 5 Compiled AI Directive) to prevent hallucinations, while **Circuit Breakers** monitor Mean Time Between Hallucinations (MTBH)[cite: 2].
- **Simplex Action Gates:** Destructive or legally binding operations (e.g., executing a multi-million-dollar tax remittance or terminating benefits) trigger Simplex Action Gates that halt autonomous execution for human-in-the-loop sign-off[cite: 2].

Here is the detailed technical walkthrough for the **5D Requirement Context Bundle (RCB) Traversal Flow** and the **SHACL-to-Policy Compilation Pipeline**, applied to Insperity's **Payroll & Tax Administration** domain.

Part 1: 5D Requirement Context Bundle (RCB) Traversal Flow

When an SMB client administrator initiates a multi-state payroll run, the request enters the **L3 Gateway Holon (RDSG)** as a structured transaction object. Instead of passing directly to execution scripts, it undergoes a 5-dimensional context-aware holarchic traversal.

[Inbound Request] ──► [Stage 0: 5D RCB Assembly] ──► [Steps 1–4: Enforcement Engine] ──► [Step 5: Compiled Directive] ──► [Simplex Gate]

Scenario

A client payroll admin executes gross-to-net processing for employees working across California and federal jurisdictions with overtime hours.

Stage 0: 5D RCB Context Deconstruction

The gateway deconstructs the request into its 5-dimensional context vector to bind the transaction to the correct domain ontologies and policy sets:

Relevancy Dimension	Vector Payload	Holonic Role & Resolution PDF
WHO	Caller: ClientAdmin_4412 Tenant: SMB_Client_AcmeCorp Target: Worker_CA_8819	Resolves persona access scope, verifies RBAC/ABAC entitlements, and confirms multi-tenant isolation boundaries.
WHAT	isp-pay:TimesheetRecord isp-pay:GrossToNetBatch	Activates canonical OWL classes, SHACL invariant shapes, and tax calculation concepts.
WHEN	PeriodEnd: 2026-08-15 TaxYear: 2026	Activates versioned SWRL temporal tax rules, 2026 SUTA/FUTA wage ceilings, and KGCL timestamps.
WHERE	Jurisdiction: US-CA Domain: isp-domain:Payroll	Routes to the sovereign Payroll & Tax Domain Holon and federates jurisdiction-specific policy sets.

WHY	Intent: GrossToNetExecution Mandate: CADailyOvertime_Sec510	Activates LKIF deontic obligations (statutory mandatory overtime payment) and audit logging constraints.

Steps 1–4: Enforcement Engine (Holarchic Traversal)

1. **Step 1 — Invariant Validation (SHACL):** Validates the timesheet payload against isp-pay:CADailyTimesheetShape (e.g., non-exempt status, daily hours > 8.0, correct pay rate multiplier).
2. **Step 2 — Temporal & Nexus Inference (SWRL):** Evaluates tax reciprocity rules between employee home and work jurisdictions for the 2026 tax year.
3. **Step 3 — PEP Compliance Evaluation:** Checks Policy Enforcement Points across **Tax & Payroll PEP** (SUTA rate verification) and **Security PEP** (banking encryption check).
4. **Step 4 — STPA/STAMP Safety Verification:** Verifies Unsafe Control Actions (UCAs)—e.g., ensuring no negative wage deductions or double pay disbursements occur.

Step 5: Compiled Directive & Delivery Gate

- **Directive Compilation:** Emits a deterministic execution directive containing the verified gross-to-net parameters and statutory withholding matrices.
- **Simplex Action Gate:** If the batch tax total exceeds automated clearance thresholds (e.g., $> \$500,000$), the gate suspends automated fund disbursement, generating an alert for human-in-the-loop authorization.

Part 2: SHACL-to-Policy Compilation Pipeline

To achieve microsecond execution without querying the triple store for millions of daily timesheet calculations, canonical SHACL/SWRL invariants are compiled into runtime-native targets (dbt/SQL, OPA/Rego).

```
[Canonical Invariant (RDF/SHACL)]
           │
           ▼
   1. Shape Parser
           │
           ▼
 2. Constraint Extractor
           │
           ▼
 3. Harmonization Check (SPARQL Conflict Scan)
           │
           ▼
   4. Target Mapper
           │
```

```
          ▼
 5. Policy Generator (Emits dbt/SQL / OPA Rego)
          │
          ▼
6. KGCL Version & Publish (Persistent URI)
```

Step 1: Canonical Source Invariant (Design-Time RDF/Turtle)

Defined in the **Payroll & Tax Domain Holon (isp-pay)**:

Code snippet

```
@prefix isp-core: <https://insperity.com/ontologies/core#> .
@prefix isp-pay:  <https://insperity.com/ontologies/payroll#> .
@prefix lkif:     <http://www.estrellaproject.org/lkif-core/norm.owl#> .
@prefix sh:       <http://www.w3.org/ns/shacl#> .
@prefix xsd:      <http://www.w3.org/2001/XMLSchema#> .

# Canonical LKIF Deontic Norm
isp-pay:CADailyOvertimeNorm a lkif:Obligation ;
  lkif:isSubjectTo "California Labor Code Section 510" ;
  isp-core:jurisdiction "US-CA" ;
  isp-core:workerClass isp-core:NonExempt ;
  isp-pay:dailyThresholdHours "8.0"^^xsd:decimal ;
  isp-pay:rateMultiplier "1.5"^^xsd:decimal ;
  isp-pay:excessDailyThresholdHours "12.0"^^xsd:decimal ;
  isp-pay:excessRateMultiplier "2.0"^^xsd:decimal .

# Canonical SHACL Invariant Shape
isp-pay:CADailyOvertimeShape a sh:NodeShape ;
  sh:targetClass isp-pay:DailyTimesheetEntry ;
  sh:property [
    sh:path isp-pay:dailyHours ;
    sh:datatype xsd:decimal ;
    sh:minInclusive 0.0 ;
    sh:maxInclusive 24.0 ;
  ] ;
  sh:sparql [
    a sh:SPARQLConstraint ;
    sh:message "Non-exempt California timesheets exceeding 8 hours must apply 1.5x OT." ;
    sh:select """
      PREFIX isp-pay: <https://insperity.com/ontologies/payroll#>
      PREFIX isp-core: <https://insperity.com/ontologies/core#>
      SELECT $this
      WHERE {
        $this isp-pay:jurisdiction "US-CA" ;
```

```
            isp-pay:workerClass isp-core:NonExempt ;
            isp-pay:dailyHours ?hours ;
            isp-pay:appliedOvertimeRate ?rate .
        FILTER (?hours > 8.0 && ?hours <= 12.0 && ?rate < 1.5)
      }
    """ ;
  ] .
```

Step 2: The 6-Stage Compilation Walkthrough

1. **Shape Parser:** Ingests the parsed constraint tree from isp-pay:CADailyOvertimeShape and isp-pay:CADailyOvertimeNorm.
2. **Constraint Extractor:** Extracts parameter variables (jurisdiction = "US-CA", workerClass = "NonExempt", tier1_hours = 8.0, tier1_mult = 1.5, tier2_hours = 12.0, tier2_mult = 2.0).
3. **Harmonization Check:** Runs a SPARQL conflict scan to ensure state-level daily overtime rules correctly override federal weekly-only 40-hour thresholds:
4. Code snippet

```
PREFIX isp-core: <https://insperity.com/ontologies/core#>
PREFIX isp-pay: <https://insperity.com/ontologies/payroll#>

SELECT ?rule ?jurisdiction ?priority
WHERE {
  ?rule a lkif:Obligation ;[cite: 2]
      isp-core:jurisdiction ?jurisdiction ;
      isp-pay:precedenceLevel ?priority .
  FILTER (?jurisdiction IN ("US-FED", "US-CA"))
}
ORDER BY DESC(?priority)
```

5.
6.
7. **Target Mapper:** Determines the target execution engines:
   - **Engine A (Data Pipeline / Batch Payroll):** Target = dbt_sql_test [cite: 2]
   - **Engine B (API / Policy Gateway):** Target = opa_rego_policy [cite: 2]
8. **Policy Generator:** Emits the runtime artifacts (shown below)[cite: 2].
9. **KGCL Version & Publish:** Stamps the generated code with a unique KGCL version and metadata linking back to isp-pay:CADailyOvertimeShape[cite: 2].

Step 3: Emitted Runtime Policy Artifacts

Target A: Generated dbt / SQL Check Constraint (Batch Gross-to-Net Validation)

URI Traceability:
https://insperity.com/ontologies/payroll#CADailyOvertimeShape

[cite: 2]

SQL
```
-- Compiled by Rule Projection Agent from isp-pay:CADailyOvertimeShape
-- KGCL Record: KGCL_2026_PAY_CA_0817

SELECT
    timesheet_id,
    employee_id,
    client_id,
    work_date,
    daily_hours,
    applied_overtime_multiplier,
    'VIOLATION_CA_DAILY_OT_RATE_INVALID' AS error_code,
    'https://insperity.com/ontologies/payroll#CADailyOvertimeShape' AS source_invariant_uri
FROM {{ ref('fct_daily_timesheet_entries') }}
WHERE jurisdiction = 'US-CA'
  AND worker_classification = 'NON_EXEMPT'
  AND (
      (daily_hours > 8.0 AND daily_hours <= 12.0 AND applied_overtime_multiplier < 1.5)
      OR
      (daily_hours > 12.0 AND applied_overtime_multiplier < 2.0)
  );
```

Target B: Generated OPA / Rego Policy (Real-Time API Gateway Authorization)
URI Traceability:
https://insperity.com/ontologies/payroll#CADailyOvertimeNorm

[cite: 2]

Code snippet
```
package insperity.payroll.compliance

# Compiled by Rule Projection Agent from isp-pay:CADailyOvertimeNorm
# Source URI: https://insperity.com/ontologies/payroll#CADailyOvertimeNorm

default calculate_overtime_multiplier = 1.0

# Tier 1: CA Daily Overtime (8 to 12 Hours)
calculate_overtime_multiplier = 1.5 {
```

```
    input.jurisdiction == "US-CA"
    input.worker_classification == "NON_EXEMPT"
    input.daily_hours > 8.0
    input.daily_hours <= 12.0
}

# Tier 2: CA Daily Double Time (> 12 Hours)
calculate_overtime_multiplier = 2.0 {
    input.jurisdiction == "US-CA"
    input.worker_classification == "NON_EXEMPT"
    input.daily_hours > 12.0
}

# Enforce compliance validation gate
deny_payroll_entry[msg] {
    input.jurisdiction == "US-CA"
    input.worker_classification == "NON_EXEMPT"
    input.daily_hours > 8.0
    input.claimed_multiplier < calculate_overtime_multiplier
    msg := sprintf(
        "Invalid overtime multiplier %v for %v hours in CA. Required: %v (Source URI: %v)",
        [input.claimed_multiplier, input.daily_hours, calculate_overtime_multiplier,
"https://insperity.com/ontologies/payroll#CADailyOvertimeNorm"]
    )
}
```

Part 3: Architecture Value for Insperity

- **Zero Graph Latency at Runtime:** Runtime evaluation takes sub-milliseconds in OPA or dbt/SQL pipelines without invoking graph reasoners on millions of client records[cite: 2].
- **Single Source of Truth (SSOT):** If California or federal overtime laws change, legal architects update the single canonical OWL/SHACL triple in the isp-pay ontology; the compilation pipeline automatically propagates the compiled policy across all database engines and microservices[cite: 2].
- **Complete Audit Traceability:** Every rejected timesheet or calculated wage line item carries an immutable URI back to the specific statutory LKIF and SHACL definition that governed it[cite: 2].